

Simplicial Homotopy Theory

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Abstract

References:

Contents

1	Simplicial Homotopy Theory	3
1.1	Homotopy of Simplicial Sets	3
1.2	Homotopic Morphisms	4
1.3	Homotopy Equivalences	4
2	Fibrations and Cofibrations	6
2.1	Lifting Properties	6
2.2	Kan Fibrations	6
2.3	Anodyne Morphisms	6
2.4	Trivial Kan Fibrations	7
3	The Simplicial Homotopy Groups	9
3.1	The Simplicial Homotopy Groups	9
3.2	Weak Homotopy Equivalences	9
3.3	Kan Complexes as Spaces	10
3.4	Contractibility	11
4	Fibrant Replacement	12
4.1	Factorization of Morphisms into (Co)Fibrations	12
4.2	Ex^∞ and the Subdivision Functor	12
4.3	Fibrant Replacements	12
5	Homotopy Limits and Colimits	13
5.1	The Standard Homotopy Pullback and Pushout	13
5.2	Homotopy Pullback and Pushout Squares	13
5.3	Recognizing Homotopy Pullbacks and Pushouts	13

1 Simplicial Homotopy Theory

Algebraic topology is a new subject which received a name change from combinatorial topology. At that time, combinatorics was highly involved in deriving topological invariants because the combinatorial structure of any invariants or spaces of study makes computation easy. In particular, a great deal of work has been put into the homotopy theory of simplicial spaces.

Nowadays, the study of combinatorial objects in topology is less prominent, but the category of simplicial sets still play a distinguished role in algebraic topology. Indeed the category of simplicial sets is the prototypical example of a model category (we have not seen it), as well as exhibiting a Quillen equivalence between the model of a simplicial set and some model category structure of **Top**. We will not explore the concept of model categories but will instead display foundational knowledge of it in order to develop the category of simplicial sets as a workable category when studying Model Categories. This means that we will develop the notion of homotopy, fibrations and cofibrations in this category.

1.1 Homotopy of Simplicial Sets

Definition 1.1.1 (Homotopy between Two Morphisms) Let X, Y be simplicial sets. Let $f, g : X \rightarrow Y$ be two morphisms. A simplicial homotopy from f to g is a morphism $\eta : X \times \Delta^1 \rightarrow Y$ such that the following diagram commutes:

$$\begin{array}{ccccc}
 X \simeq X \times \Delta^0 & \xrightarrow{\text{id}_X \times \delta_1} & X \times \Delta^1 & \xleftarrow{\text{id}_X \times \delta_0} & X \simeq X \times \Delta^0 \\
 & \searrow f & \downarrow \exists \eta & \swarrow g & \\
 & & Y & &
 \end{array}$$

If there is a homotopy from f to g , then in general there may not be a homotopy from g to f . This is different from classical homotopy theory. However, recall that $\text{Sing}(X)$ is a Kan complex for every space X . We will see that at least the homotopy theory of Kan complexes is the same as that of topological spaces. In particular, we will prove that for Kan complexes, there exists a homotopy from f to g if and only if there exists a homotopy from g to f .

Proposition 1.1.2 Let X, Y be simplicial sets. Let $f, g : X \rightarrow Y$ be morphisms. Then there exists a homotopy from g to f if and only if there exists a family of morphisms $h_i^n : X_n \rightarrow Y_{n+1}$ such that the following are true.

- $d_0^n \circ h_0 = f_n$
- $d_0^{n+1} \circ h_1 = g_n$
- The composition

$$d_i \circ h_j = \begin{cases} h_{j-1} \circ d_i^{n+1} & \text{if } 0 \leq i < j \leq n \\ d_i \circ h_{i-1} & \text{if } i = j \neq 0 \\ h_j \circ d_{i-1} & \text{if } i > j + 1 \end{cases}$$

- The composition

$$s_i \circ h_j = \begin{cases} h_{j+1} \circ s_i & \text{if } i \leq j \\ h_j \circ s_{i-1} & \text{if } i > j \end{cases}$$

Definition 1.1.3 (Homotopy Relative to Subsets) Let X, Y be simplicial sets. Let $K \subseteq X$ be a simplicial subset. Let $\iota : K \hookrightarrow X$ denote the inclusion. Let $f, g : X \rightarrow Y$ be two morphisms. Let $\eta : X \times \Delta^1 \rightarrow Y$ be a homotopy from f to g . We say that ι is a homotopy relative to K if the following diagram commutes:

$$\begin{array}{ccc}
K \times \Delta^1 & \xrightarrow{\text{proj}_K} & K \\
\downarrow \iota \times \text{id}_{\Delta^1} & & \downarrow \alpha \\
X \times \Delta^1 & \xrightarrow{\eta} & Y
\end{array}$$

where $\alpha = f|_K = g|_K$.

Notice that if $K = \emptyset$ then homotopies relative to a subset is just a homotopy without restrictions.

Definition 1.1.4 (Pointed Homotopies)

Let $(X, x_0), (Y, y_0)$ be pointed simplicial sets. Let $f, g : (X, x_0) \rightarrow (Y, y_0)$ be pointed simplicial maps. A pointed homotopy from f to g is a homotopy from f to g relative to $\{x_0\}$.

1.2 Homotopic Morphisms

Definition 1.2.1 (Homotopic Morphisms)

Let S, T be simplicial sets. Let $f, g : S \rightarrow T$ be morphisms. Let $K \subseteq S$ be a simplicial subset. We say that f and g are homotopic relative to K if there exists a sequence of morphisms $f = f_0, \dots, f_n = g$ such that for $0 \leq i \leq n-1$, either there is a homotopy from f_{i-1} to f_i relative to K or there is a homotopy from f_i to f_{i-1} relative to K .

If we replace S and T with topological spaces in the above definition, then the notion of homotopic morphisms coincide with that of **Top** because every homotopy can be reversed.

Proposition 1.2.2

Let S, T be simplicial sets. Let $f, g : S \rightarrow T$ be morphisms. Then f and g are homotopic if and only if $[f] = [g] \in \pi_0(\mathbf{Hom}_{\mathbf{sSet}}(S, T))$.

Proposition 1.2.3

Let S, T be Kan complexes. Let $f, g : S \rightarrow T$ be morphisms. Then the following are equivalent.

- f and g are homotopic.
- $[f] = [g] \in \pi_0(\mathbf{Hom}_{\mathbf{sSet}}(S, T))$.
- There exists a homotopy from f to g .

1.3 Homotopy Equivalences

Definition 1.3.1 (Simplicial Homotopy Inverse)

Let S, T be simplicial sets. Let $f : S \rightarrow T$ and $g : T \rightarrow S$ be maps of simplicial sets. We say that g is a simplicial homotopy inverse of f if $g \circ f$ is homotopic to id_S and $f \circ g$ is homotopic to id_T .

Lemma 1.3.2

Let S, T be simplicial sets. Let $f : S \rightarrow T$ be a morphism. Suppose that $g, h : T \rightarrow S$ are simplicial homotopy inverses of f . Then g and h are homotopic.

Definition 1.3.3 (Homotopy Equivalences)

Let S, T be simplicial sets. Let $f : S \rightarrow T$ be a map of simplicial sets. We say that f is a homotopy equivalence if f admits a simplicial homotopy inverse.

Lemma 1.3.4

Let S, T be simplicial sets. Let $f : S \rightarrow T$ be a homotopy equivalence. If $g : S \rightarrow T$ is homotopic to f , then g is a homotopy equivalence.

Proposition 1.3.5

Let S, T be simplicial sets. Let $f : S \rightarrow T$ be a morphism. Then the following are equivalent.

- f is a homotopy equivalence.
- For every simplicial set U , post composition induces a bijection

$$\pi_0(\mathrm{Hom}_{\mathbf{sSet}}(U, X)) \rightarrow \pi_0(\mathrm{Hom}_{\mathbf{sSet}}(U, Y))$$

- For every simplicial set U , pre composition induces a bijection

$$\pi_0(\mathrm{Hom}_{\mathbf{sSet}}(Y, U)) \rightarrow \pi_0(\mathrm{Hom}_{\mathbf{sSet}}(X, U))$$

Proposition 1.3.6

Let S, T, U be simplicial sets. Let $f : S \rightarrow T$ and $g : T \rightarrow U$ be morphisms. If any two of $f, g, g \circ f$ are homotopy equivalence, then third one is also a homotopy equivalence.

Proposition 1.3.7

Let $(X, x_0), (Y, y_0)$ be pointed Kan complexes. Let $f : (X, x_0) \rightarrow (Y, y_0)$ be a map of pointed simplicial sets. Then f is a homotopy equivalence if and only if $f : X \rightarrow Y$ is a homotopy equivalence.

Definition 1.3.8 (The Homotopy Category of Kan Complexes)

Define the homotopy category $h\mathbf{Kan}$ of Kan complexes as follows.

- The objects are Kan complexes.
- The morphisms are given by homotopy classes $[f]$ for a morphism $f : X \rightarrow Y$ of Kan complexes. In other words, $\mathrm{Hom}_{h\mathbf{Kan}}(X, Y) = \pi_0(\mathrm{Hom}_{\mathbf{sSet}}(X, Y))$.
- Composition is given by the formula $[g] \circ [f] = [g \circ f]$

2 Fibrations and Cofibrations

2.1 Lifting Properties

Definition 2.1.1 (The Left Lifting Property) Let $A, B, X, Y \in \mathbf{sSet}$ be simplicial sets. Let $p : X \rightarrow Y$ and $i : A \rightarrow B$ be morphisms. We say that i has the left lifting property with respect to p if the following is true. For all morphisms $f : A \rightarrow X$ and $g : B \rightarrow Y$ such that $p \circ f = g \circ i$, there exists $h : A \rightarrow Y$ such that the following diagram commutes:

$$\begin{array}{ccc} A & \xrightarrow{f} & X \\ i \downarrow & \nearrow \exists h & \downarrow p \\ B & \xrightarrow{g} & Y \end{array}$$

Definition 2.1.2 (The Right Lifting Property) Let $A, B, X, Y \in \mathbf{sSet}$ be simplicial sets. Let $p : X \rightarrow Y$ and $i : A \rightarrow B$ be morphisms. We say that p has the right lifting property with respect to i if the following is true. For all morphisms $f : A \rightarrow X$ and $g : B \rightarrow Y$ such that $p \circ f = g \circ i$, there exists $h : B \rightarrow X$ such that the following diagram commutes:

$$\begin{array}{ccc} A & \xrightarrow{f} & X \\ i \downarrow & \nearrow \exists h & \downarrow p \\ B & \xrightarrow{g} & Y \end{array}$$

2.2 Kan Fibrations

Definition 2.2.1 (Kan Fibrations) Let $f : X \rightarrow Y$ be a morphism of simplicial sets. We say that f is a Kan fibration if the following condition is satisfied: For every commutative diagram:

$$\begin{array}{ccc} \Lambda_k^n & \longrightarrow & X \\ \downarrow & & \downarrow f \\ \Delta^n & \longrightarrow & Y \end{array}$$

where $n \geq 1$ and $0 \leq k \leq n$, there exists a lift $\Delta^n \rightarrow X$ such that the following diagram commutes:

$$\begin{array}{ccc} \Lambda_k^n & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow f \\ \Delta^n & \longrightarrow & Y \end{array}$$

Lemma 2.2.2 Let X be a simplicial set. Then X is a Kan complex if and only if the unique map $X \rightarrow *$ is a Kan fibration.

Proposition 2.2.3

Let X, Y be simplicial sets. Let $f : X \rightarrow Y$ be a Kan fibration. If Y is a Kan complex, then X is a Kan complex.

Closed under retracts, pullbacks, filtered colimits, products with standard simplicies, composition

2.3 Anodyne Morphisms

They are the acyclic cofibrations of $\mathbf{sSet}$ with standard model structure.

Definition 2.3.1 (Anodyne Morphisms) Let $X, Y \in \mathbf{sSet}$ be simplicial sets. Let $f : X \rightarrow Y$ be a morphism. We say that f is anodyne if f has the left lifting property with respect to all Kan fibrations:

$$\begin{array}{ccc} A & \longrightarrow & X \\ \text{Kan Fib.} \downarrow & \nearrow \exists h & \downarrow f \\ B & \longrightarrow & Y \end{array}$$

2.4 Trivial Kan Fibrations

They are the acyclic fibrations of $\mathbf{sSet}$ with standard model structure.

Definition 2.4.1 (Trivial Kan Fibrations) Let $X, Y \in \mathbf{sSet}$ be simplicial sets. Let $f : X \rightarrow Y$ be a morphism. We say that f is a trivial Kan fibration if f has the right lifting property with respect to all inclusions $\partial\Delta^n \hookrightarrow \Delta^n$:

$$\begin{array}{ccc} \partial\Delta^n & \longrightarrow & X \\ \downarrow & \nearrow \exists k & \downarrow f \\ \Delta^n & \longrightarrow & Y \end{array}$$

Proposition 2.4.2 Let $X, Y \in \mathbf{sSet}$ be simplicial sets. Let $f : X \rightarrow Y$ be a morphism. Then f is a trivial Kan fibration if and only if f has the right lifting property with respect to all monomorphisms $i : A \rightarrow B$ of simplicial sets:

$$\begin{array}{ccc} A & \xrightarrow{f} & X \\ \downarrow & \nearrow g & \\ B & & \end{array}$$

Corollary 2.4.3 Let $X, Y \in \mathbf{sSet}$ be simplicial sets. Let $f : X \rightarrow Y$ be a morphism. If f is a trivial Kan fibration, then f admits a section. This means that there exists a morphism $s : Y \rightarrow X$ such that

$$f \circ s = \text{id}_Y$$

Moreover, $s \circ f : X \rightarrow X$ is homotopic to the identity map id_X .

Corollary 2.4.4

Let X, Y be simplicial sets. Let $f : X \rightarrow Y$ be trivial Kan fibration. Then f is a homotopy equivalence.

Proposition 2.4.5

Let X, Y be simplicial sets. Let $f : X \rightarrow Y$ be map of simplicial sets. Then f is a trivial Kan fibration if and only if f is a weak homotopy equivalence and a Kan fibration.

Proposition 2.4.6 Let $X, Y \in \mathbf{sSet}$ be simplicial sets. Let $f : X \rightarrow Y$ be a trivial Kan fibration. If X is a Kan complex, then Y is a Kan complex.

Proposition 2.4.7 Let $X, Y \in \mathbf{sSet}$ be simplicial sets. Let $f : X \rightarrow Y$ be a morphism. If f is a trivial Kan fibration, then the induced morphism

$$f \circ - : \text{Hom}_{\mathbf{sSet}}(A, X) \rightarrow \text{Hom}_{\mathbf{sSet}}(A, Y)$$

is a trivial Kan fibration for all $A \in \mathbf{sSet}$.

3 The Simplicial Homotopy Groups

3.1 The Simplicial Homotopy Groups

Because homotopies from X to Y is an equivalence relation only when Y is a Kan complex, we can only define simplicial homotopy groups for Kan complexes. This notion can easily be extended to arbitrary simplicial sets by replacing the simplicial set with a fibrant replacement (and hence becomes a Kan complex).

Definition 3.1.1 (Simplicial Homotopy Groups) Let X be a Kan complex. Let $v \in X$ be a vertex of X . Define the simplicial homotopy groups of (X, v) as follows:

- For $n \geq 1$, define the n th simplicial homotopy group $\pi_n^{\text{sSet}}(X, v)$ of X at v to be the set of homotopy classes of maps $[\alpha : \Delta^n \rightarrow X]$ relative to boundary $\partial\Delta^n$. In other words, the following diagram commutes:

$$\begin{array}{ccc} \partial\Delta^n & \xrightarrow{!} & \Delta^0 \\ \downarrow & & \downarrow v \\ \Delta^n & \xrightarrow{\alpha} & X \end{array}$$

- Define the 0th simplicial homotopy group $\pi_0^{\text{sSet}}(X)$ of X to be the set of homotopy classes of vertices of X .

If X is a general simplicial set, define the simplicial homotopy group of X to be

$$\pi_n^{\text{sSet}}(X, v) = \pi_n^{\text{sSet}}(PX, Pv)$$

where PX is a fibrant replacement of X .

Theorem 3.1.2 Let X be a Kan complex. Let $f, g : \Delta^n \rightarrow X$ be two representatives of two elements in $\pi_n(X, v)$ for $n \geq 1$. Then the following data

$$v_i = \begin{cases} s_0^n(v) & \text{if } 0 \leq i \leq n-2 \\ f & \text{if } i = n-1 \\ g & \text{if } i = n+1 \end{cases}$$

defines a horn $\Lambda_i^{n+1} \rightarrow X$. Such a map extends to a map $\theta : \Delta^{n+1} \rightarrow X$. Define

$$[f] \cdot [g] = d_n \theta$$

Then such an operation is well defined on the equivalence class. Moreover, it defines a group operation on $\pi_n^{\text{sSet}}(X, v)$.

Functoriality, homotopy invariance, independence of base point (on connected components)

Proposition 3.1.3 Let X be a Kan complex. Then for $n \geq 2$, the above group structure on $\pi_n^{\text{sSet}}(X, v)$ is abelian.

3.2 Weak Homotopy Equivalences

Definition 3.2.1 (Weak Homotopy Equivalence)

Let X, Y be simplicial sets. Let $f : X \rightarrow Y$ be a map of simplicial sets. We say that f is a weak homotopy equivalence if f induces isomorphisms

$$\pi_n^{\text{sSet}}(X, x) \cong \pi_n^{\text{sSet}}(Y, f(x))$$

for all $x \in X$ and all $n \in \mathbb{N}$.

Let X, Y be simplicial sets. The existence of a weak homotopy equivalence $f : X \rightarrow Y$ does not guarantee the existence of a weak homotopy equivalence $g : Y \rightarrow X$.

Proposition 3.2.2

Let X, Y be simplicial sets. Let $f : X \rightarrow Y$ be a map of simplicial sets. Then f is a weak homotopy equivalence if and only if for all simplicial sets Z , precomposition with f induces a bijection $\pi_0(\text{Hom}_{\mathbf{sSet}}(Y, Z)) \rightarrow \pi_0(\text{Hom}_{\mathbf{sSet}}(X, Z))$.

Lemma 3.2.3

Let X, Y be simplicial sets. Let $f : X \rightarrow Y$ be a map of simplicial sets. If f is a homotopy equivalence, then f is a weak homotopy equivalence.

Theorem 3.2.4 (Whitehead's Theorem for Simplicial Sets)

Let X, Y be Kan complexes. Let $f : X \rightarrow Y$ be a map of simplicial sets. Then f is a homotopy equivalence if and only if f is a weak homotopy equivalence.

Lemma 3.2.5

Let X, Y, Z be simplicial sets. Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be maps of simplicial sets. If two out of $f, g, g \circ f$ are weak homotopy equivalences, then the third one is a weak homotopy equivalence.

3.3 Kan Complexes as Spaces

Proposition 3.3.1

Let X be a simplicial set. Let $v \in X_0$. Then there is an isomorphism

$$\pi_n^{\mathbf{sSet}}(X, v) \cong \pi_n(|X|, v)$$

for all $n \in \mathbb{N}$.

Proposition 3.3.2

Let X be a space. Let $x \in X$. Then there is an isomorphism

$$\pi_n(X, x) \cong \pi_n^{\mathbf{sSet}}(\text{Sing}(X), x)$$

Corollary 3.3.3

Let X be a space. Let $x \in X$. Then the counit map $|\text{Sing}(X)| \rightarrow X$ is a weak homotopy equivalence.

Corollary 3.3.4

Let X, Y be simplicial sets. Let $f : X \rightarrow Y$ be a map of simplicial sets. Then f is a weak homotopy equivalence if and only if $|f| : |X| \rightarrow |Y|$ is a weak homotopy equivalence.

Since every space is a Kan complex, this means that the notion of weak equivalence in Kan complexes is the same as the notion of weak equivalence in spaces. Thus when one only cares about weak homotopy type, **Kan** subsumes **Top**. By Whitehead's theorem, when one only cares about homotopy types, then **Kan** subsumes **CW**.

Proposition 3.3.5

Let S be a Kan complex. Then the unit map $S \rightarrow \text{Sing}(|S|)$ is a homotopy equivalence.

Corollary 3.3.6

The geometric realization functor induces an equivalence of categories

$$h\mathbf{Kan} \simeq h\mathbf{CW}$$

3.4 Contractibility

Definition 3.4.1 (Contractible Simplicial Sets)

Let X be a simplicial set. We say that X is contractible if the unique map $X \rightarrow *$ is a homotopy equivalence.

Proposition 3.4.2 Let X be a simplicial set. Then the following are equivalent.

- X is a contractible Kan complex (every morphism $\partial\Delta^n \rightarrow X$ has an extension $\Delta^n \rightarrow X$)
- The unique map $X \rightarrow \Delta^0$ is a trivial Kan fibration.
- The unique map $X \rightarrow *$ has the right lifting property with respect to all monomorphisms $i : A \rightarrow B$ of simplicial sets:

$$\begin{array}{ccc} A & \xrightarrow{f} & X \\ \downarrow & \nearrow g & \\ B & & \end{array}$$

Proposition 3.4.3

Let X be a Kan complex. Then the following are equivalent.

- X is contractible.
- $\pi_n(X, x_0) = 0$ for all $x_0 \in X$ and all $n \in \mathbb{N}$.
- $X \rightarrow *$ is a trivial Kan fibration.
- The unique map $X \rightarrow *$ has the right lifting property with respect to all monomorphisms $i : A \rightarrow B$ of simplicial sets:

$$\begin{array}{ccc} A & \xrightarrow{f} & X \\ \downarrow & \nearrow g & \\ B & & \end{array}$$

Proposition 3.4.4 Let $X, Y \in \mathbf{sSet}$ be simplicial sets. Let $f : X \rightarrow Y$ be a trivial Kan fibration. If X is a contractible Kan complex, then Y is a contractible Kan complex.

Definition 3.4.5 (Weakly Contractible)

Let X be a simplicial set. We say that X is weakly contractible if the unique map $X \rightarrow *$ is a weak homotopy equivalence.

Clearly if X is contractible, then it is weakly contractible. By Whitehead's theorem, if X is a Kan complex, the converse also holds.

4 Fibrant Replacement

4.1 Factorization of Morphisms into (Co)Fibrations

Proposition 4.1.1

Let X, Y be simplicial sets. Let $f : X \rightarrow Y$ be a map of simplicial sets. Then there exists a factorization

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow f' & \nearrow f'' \\ & Q(f) & \end{array}$$

such that the following are true.

- f' is anodyne.
- f'' is a Kan fibration.
- $Q(f)$ is a Kan complex.

Proposition 4.1.2 Let X, Y be simplicial sets. Let $f : X \rightarrow Y$ be a morphism. Then there exists a choice of $Q(f)$ such that Q defines a functor

$$Q : \text{Hom}_{\mathbf{Cat}}(\Delta^1, \mathbf{sSet}) \rightarrow \mathbf{sSet}$$

that commutes with filtered colimits.

4.2 Ex^∞ and the Subdivision Functor

4.3 Fibrant Replacements

Definition 4.3.1 (Fibrant Replacements) Let X be a simplicial set. A fibrant replacement of X is a Kan complex QX such that the unique map $X \rightarrow *$ factorizes into $X \rightarrow QX \rightarrow *$ where $X \rightarrow QX$ is anodyne.

Secret: these are just the fibrant replacements in model category.

Lemma 4.3.2 Let X be a simplicial set. Then the following are all fibrant replacements of X .

- $Q_X = Q(X \rightarrow *)$
- $\text{Ex}^\infty(X)$

5 Homotopy Limits and Colimits

5.1 The Standard Homotopy Pullback and Pushout

Definition 5.1.1 (The Standard Homotopy Pullbacks) Let the following be a diagram of simplicial sets and morphisms.

$$X \xrightarrow{f} Z \xleftarrow{g} Y$$

Define the standard homotopy pullback to be the simplicial set

$$\mathrm{Holim}_Q(X \rightarrow Z \leftarrow Y) = X \times_Z^h Y = X \times_{\mathrm{Hom}_{\mathbf{sSet}}(\{0\}, Z)} \mathrm{Hom}_{\mathbf{sSet}}(\Delta^1, Z) \times_{\mathrm{Hom}_{\mathbf{sSet}}(\{1\}, Z)} Y$$

Here we identify the vertices of Δ^1 as 0 and 1, and we use the isomorphism $\mathcal{C} \cong \mathrm{Hom}_{\mathbf{sSet}}(\Delta^0, Z)$.

Kerodon 3.4.0.3

Lemma 5.1.2 Let the following be a diagram of simplicial sets and morphisms, where Z is a Kan complex.

$$X \xrightarrow{f} Z \xleftarrow{g} Y$$

Then the following diagram commutes up to homotopy:

$$\begin{array}{ccc} \mathrm{Holim}_Q(X \rightarrow Z \leftarrow Y) & \xrightarrow{\mathrm{proj}} & X \\ \mathrm{proj} \downarrow & & \downarrow f \\ Y & \xrightarrow{g} & Z \end{array}$$

5.2 Homotopy Pullback and Pushout Squares

5.3 Recognizing Homotopy Pullbacks and Pushouts

Proposition 5.3.1 (Kerodon 3.4.0.7) Let the following be a diagram in simplicial sets.

$$X \xrightarrow{f} Z \xleftarrow{g} Y$$

If Z is a Kan complex and one of f or g is a Kan fibration, then the following are true.

- The following diagram

$$\begin{array}{ccc} X \times_Z Y & \xrightarrow{\mathrm{pr}} & X \\ \mathrm{pr} \downarrow & & \downarrow f \\ Y & \xrightarrow{g} & Z \end{array}$$

is a homotopy pullback square.

- The inclusion map

$$X \times_Z Y \hookrightarrow X \times_Z^h Y$$

is a weak homotopy equivalence.

- $X \times_Z Y$ is a homotopy pullback of the diagram.

Secret: if the diagram is fibrant in the diagram category, then the homotopy pullback is the same as the standard pullback up to weak equivalence.